Hector Ovalle

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EDUCATION

Georgia Institute of Technology **May 2026 - December 2028**
Bachelor of Science in Electrical Engineering **GPA: 4.0**
Relevant Coursework: Digital System Design, Circuit Analysis, Signal Processing, Engineering Software Design
Professional Organizations: SiliconJackets, RoboJackets, Society of Hispanic Professional Engineers

EXPERIENCE

Augusta University
Computer Science Undergraduate Course Assistant **August 2025 - December 2025**

- Embedded in multiple course sections to assist students in real time, answering conceptual and coding-related questions to improve understanding and workflow.
- Guided students in using GitHub for version control and collaborative development, increasing code accessibility and ease of iteration.
- Contributed to a 34% improvement in overall student performance compared to previous semesters.

PROJECTS

32-bit RISC-V CPU - Verilog/SystemsVerilog **July 2026 - August 2026**

- Designed and implemented a 32-bit RISC-V CPU with instruction fetch, register-file interfacing, instruction decoding, and ALU logic.
- Implemented arithmetic, immediate, shift, load/store, branch, and halt instructions, including sign-extension for RISC-V immediate formats.
- Integrated instruction/data SRAM interfaces and verified processor behavior through simulation and debugging in a Linux development environment.

Daisy-Chained Sensor Suite PCB | KiCad, RP2040, I²C, SPI **2026**

- Designed an RP2040-based daisy-chainable sensor PCB integrating six I²C peripherals, SPI communication interfaces, and Micro-USB connectivity for power delivery and firmware programming.
- Developed detailed circuit schematics and multilayer PCB layouts in KiCad, incorporating power distribution, signal routing, component placement, and communication interfaces for modular sensor integration.
- Researched component datasheets and documented hardware specifications in a Hardware Reference Manual (HRM), ensuring design compatibility with JLCPCB manufacturing constraints and supporting future firmware development.

Low-Cost Prosthetics **January 2025 - May 2025**

- Collaborated with a team of four to design low-cost prosthetic devices using Arduino, Python, and OpenCV for motion detection.
- Developed a camera-based control system that tracked user movements in Python and interfaced with Arduino to drive servo motors for finger and arm motion.Optimized code and hardware communication, reducing system delay by 70%

TECHNICAL SKILLS

Programming: Python, C#, C++, Javascript, MATLAB, HTML/CSS
Hardware: PCB Design, Arduino, Oscilloscope, AC/DC power supplies,DMM, Hardware debugging
Software: KiCAD, Multisim, VS Code, Anaconda, OpenCV, GitHub, Microsoft Suite, Arduino IDE, Eclipse
HDL: Verilog/SystemVerilog